Robust Wearable Sleep Inference Under Realistic Mobile Sensing Constraints

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Abstract

Wearable devices such as smartwatches have made large-scale sleep monitoring possible outside clinical environments. However, deploying reliable sleep-stage inference directly on mobile hardware remains challenging due to real-world sensing constraints such as battery-saving duty cycling, OS scheduling gaps, and motion artifacts.

This project proposes a robust wearable sleep inference pipeline that operates directly on devices like the Apple Watch. Using the public PhysioNet Apple Watch dataset, which contains wrist accelerometer and photoplethysmography (PPG) heart-rate signals paired with gold-standard polysomnography (PSG) labels, we train a sleep-stage classifier to detect wake, light, deep, and REM sleep stages.

To simulate realistic mobile conditions, we introduce controlled degradation including heart-rate dropout, sampling-rate reduction, and motion artifact noise. We then apply lightweight robustness strategies such as signal imputation, multi-resolution feature extraction, and feature-level dropout to improve stability. The system is evaluated against baseline models using macro-F1, Cohen's kappa, inference latency, and memory footprint. Our goal is to demonstrate how robust on-device sleep inference can maintain high accuracy while operating under realistic mobile sensing constraints.

1 Introduction and Motivation

Wearable devices have rapidly expanded the possibilities of continuous health monitoring, particularly in sleep analysis. Smartwatches equipped with motion sensors and optical heart-rate monitors allow large-scale sleep monitoring outside clinical laboratories. These systems can provide valuable insights into sleep quality, circadian rhythms, and long-term health outcomes.

However, translating sleep inference algorithms from controlled research environments to real-world mobile devices presents significant challenges. Mobile sensing pipelines must operate under practical constraints such as battery-saving modes that reduce sensor sampling rates, operating system scheduling that introduces intermittent data gaps, and user motion that introduces noise and artifacts.

Most existing sleep inference models assume continuous, high-quality sensor streams. In practice, wearable data is often incomplete or noisy, which can significantly degrade model performance. As a result, designing sleep inference systems that remain reliable under realistic sensing conditions is an important systems problem.

This project addresses this challenge by building a robust wearable sleep inference pipeline using the PhysioNet Apple Watch dataset. By introducing controlled sensor degradation and evaluating lightweight robustness strategies, we aim to design a sleep inference system that maintains accuracy even under realistic mobile sensing constraints.

2 Related Work

Several studies have explored sleep-stage prediction using wearable sensor data. Walch et al. [1] demonstrated that Apple Watch accelerometer and heart-rate signals can be used to predict sleep stages with reasonable accuracy when validated against polysomnography (PSG). Their work also released a public dataset through PhysioNet [2], enabling further research in wearable sleep analysis.

However, most prior work assumes continuous high-quality sensor streams. Shen et al. [3] highlighted significant robustness gaps in mobile inertial tracking systems caused by real-world sensing variability such as packet loss, sampling gaps, and orientation drift.

On-device machine learning optimization has also been widely studied. Techniques such as DeepIoT [5] compress neural networks to run efficiently on resource-constrained sensing devices. Hardware-aware model optimization has also shown that lightweight architectures can maintain performance while reducing computational cost.

Despite these advances, limited work has systematically evaluated sleep inference pipelines under realistic mobile sensing degradation. Our project aims to fill this gap by stress-testing wearable sleep inference models under controlled sensor degradation scenarios and evaluating lightweight robustness techniques.

3 Proposed Method

Our proposed system processes wrist accelerometer and heart-rate signals from the PhysioNet Apple Watch dataset

to perform sleep-stage classification. The overall pipeline includes signal preprocessing, degradation simulation, robust feature extraction, and lightweight classification.

3.1 Preprocessing and Degradation Simulation

Raw accelerometer and PPG signals are first filtered and segmented into time windows suitable for sleep-stage analysis. To simulate real-world mobile sensing conditions, we introduce controlled degradation including:

- Heart-rate dropout to simulate packet loss or sensor failures
- Sampling-rate reduction to represent battery duty-cycling
- Motion artifact noise to represent real-world user movement

These modifications create realistic sensing conditions that mirror actual smartwatch behavior.

3.2 Robustness Strategies

To improve model stability under degraded sensing conditions, we introduce several lightweight robustness techniques:

- Last-value carry-forward and linear interpolation for missing heart-rate segments
- Multi-resolution feature extraction to handle reduced sampling rates
- Feature-level dropout during model training to improve robustness

These techniques allow the model to tolerate incomplete or noisy sensor streams while maintaining prediction accuracy.

3.3 Sleep Stage Classifier

Once robust features are extracted, a lightweight neural network classifier predicts sleep stages in real time. The model is designed to be small enough to run directly on wearable hardware, with a size of approximately 30 KB. This allows deployment on edge devices using frameworks such as TensorFlow Lite or CoreML.

4 Evaluation Plan

To evaluate the effectiveness of our approach, we compare the robust pipeline against baseline sleep inference models under varying levels of sensor degradation.

4.1 Experimental Setup

Experiments will be conducted using the PhysioNet Apple Watch dataset containing data from 31 participants. Controlled degradation will be applied to simulate real-world mobile sensing conditions. Model inference performance will be tested on mobile hardware including an Apple Watch simulator and a Pixel 5 smartphone.

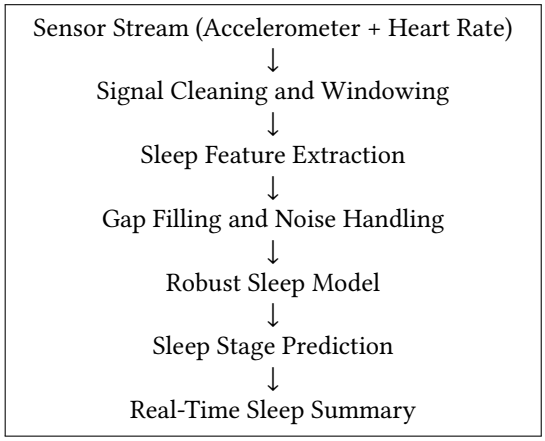

Figure 1: Overall pipeline of the proposed robust wearable sleep inference system.

4.2 Evaluation Metrics

The evaluation will focus on four key metrics:

- **Tracking Accuracy:** Macro-F1 score and Cohen's kappa for sleep-stage classification
- **Inference Speed:** Latency per inference and throughput
- **Resource Footprint:** Model size, memory usage, and energy consumption
- **Robustness Curves:** Accuracy degradation under increasing levels of missing or noisy data

4.3 Baselines

The proposed robust model will be compared against:

- The original Walch et al. sleep-stage prediction model
- Lightweight edge-based models without robustness mechanisms
- High-fidelity server-based inference pipelines

This comparison will highlight the trade-offs between model robustness, computational efficiency, and deployment feasibility on mobile hardware.

References

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